Optional Practice. Introduction to R
Quantitative Methods for Humanities

Optional enrichment for 2026–2027.
This activity is retained for students who want extra R practice. It is not part of the required continuous evaluation unless explicitly assigned by the instructor.

Understanding World Population Dynamics

Understanding population dynamics is crucial for many areas of social science. This exercise involves calculating basic demographic measures of births and deaths for the world population over two time periods: 1950–1955 and 2005–2010. The analysis will use data files `Kenya.csv`, `Sweden.csv`, and `World.csv`, which contain population data for Kenya, Sweden, and the world, respectively.

Data Description

The variables included in the datasets are described in Table 1.

Table 1: Description of Variables in the Population Data

Variable	**Description**
`country`	Abbreviated country name
`period`	Time period for data collection
`age`	Age group
`births`	Number of births (in thousands)
`deaths`	Number of deaths (in thousands)
`py.men`	Person-years for men (in thousands)
`py.women`	Person-years for women (in thousands)

The data are collected for a five-year period, with person-years representing the time contribution of each individual during the period. For example, a person who lived through the entire five-year period contributes 5 person-years, while someone who lived only half the period contributes 2.5 person-years.

1. **Crude Birth Rate (CBR):** The crude birth rate (CBR) is defined as:

$$\text{CBR} = \frac{\text{Number of births}}{\text{Number of person-years lived}}.$$

Compute the CBR for each period separately for Kenya, Sweden, and the world.

(a) Start by calculating the total person-years as the sum of person-years for men and women.

(b) Store the results as a vector of length 2 (one for each period) for each region with appropriate labels.

Briefly describe any patterns observed in the resulting CBRs.

2. **Total Fertility Rate (TFR):** The total fertility rate (TFR) adjusts for age composition in the female population. First, calculate the age-specific fertility rate (ASFR), defined as:

$$\text{ASFR}[x, x+\delta) = \frac{\text{Number of births to women of age } [x, x+\delta)}{\text{Person-years lived by women of age } [x, x+\delta)}.$$

Here, x is the starting age, and δ is the width of the age range (e.g., 5 years).

(a) Compute the ASFR for Kenya, Sweden, and the world for each period.

(b) Using the ASFR, calculate the TFR as:

$$\text{TFR} = \sum_{\text{Age Groups}} \text{ASFR}[x, x+\delta) \times \delta.$$

Store the resulting TFRs for each region and period as vectors of length 2. Describe any patterns in reproduction across the regions and periods.

3. **Crude Death Rate (CDR):** The crude death rate (CDR) is defined as:

$$\text{CDR} = \frac{\text{Number of deaths}}{\text{Number of person-years lived}}.$$

Compute the CDR for each period and region, and describe the patterns observed.

4. **Age-Specific Death Rate (ASDR):** Calculate the age-specific death rate (ASDR) for each age group and region during the 2005–2010 period:

$$\text{ASDR}[x, x+\delta) = \frac{\text{Number of deaths for people of age } [x, x+\delta)}{\text{Person-years of people of age } [x, x+\delta)}.$$

Briefly describe the patterns observed in the ASDRs.

5. **Counterfactual CDR:** To understand differences in the CDR between Kenya and Sweden, compute the counterfactual CDR for Kenya using Sweden's age distribution:

$$\text{CDR} = \sum_{\text{Age Groups}} \text{ASDR}[x, x+\delta) \times P[x, x+\delta),$$

where $P[x, x+\delta)$ is the proportion of the population in the age range $[x, x+\delta)$, calculated as the ratio of person-years in that age range to the total person-years. Compare the counterfactual CDR with the observed CDR for Kenya, and provide a substantive interpretation.